CDA 106
Numerical Methods for Data Science

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Numerical Integration

Consider

$$\int e^{-x^2}\,dx \qquad\qquad \int \frac{e^{x}}{x}\,dx$$

I $\qquad\qquad$ II

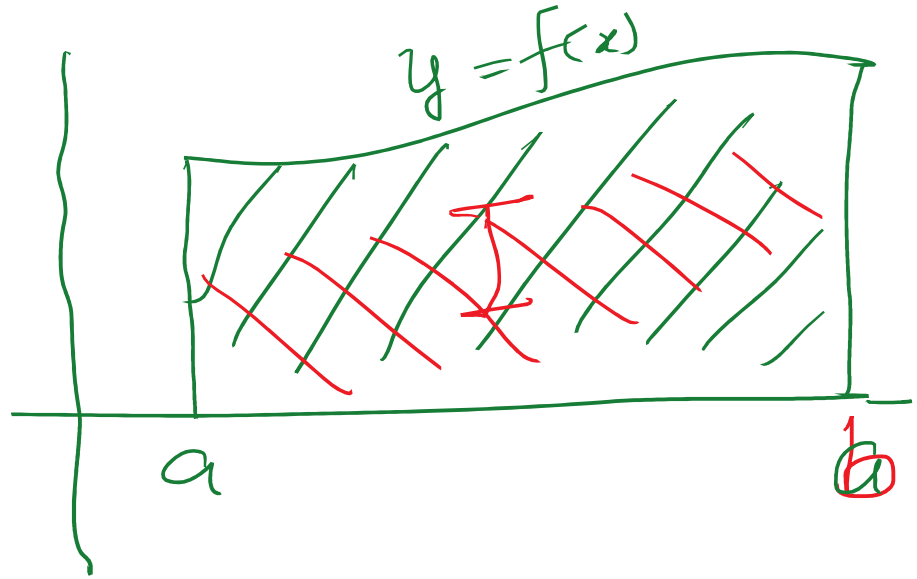
you cannot find these integrals in terms of known functions.

Integral

$$\int_a^b f(x)\,dx = \int_a^b F'(x)\,dx = F(b) - F(a)$$

$$F'(x) = f(x)$$

$$I = \int_a^b f(x)\,dx$$



x	x_0	x_1	x_2	x_3	x_4	x_5
y	y_0	y_1	y_2	y_3	y_4	y_5

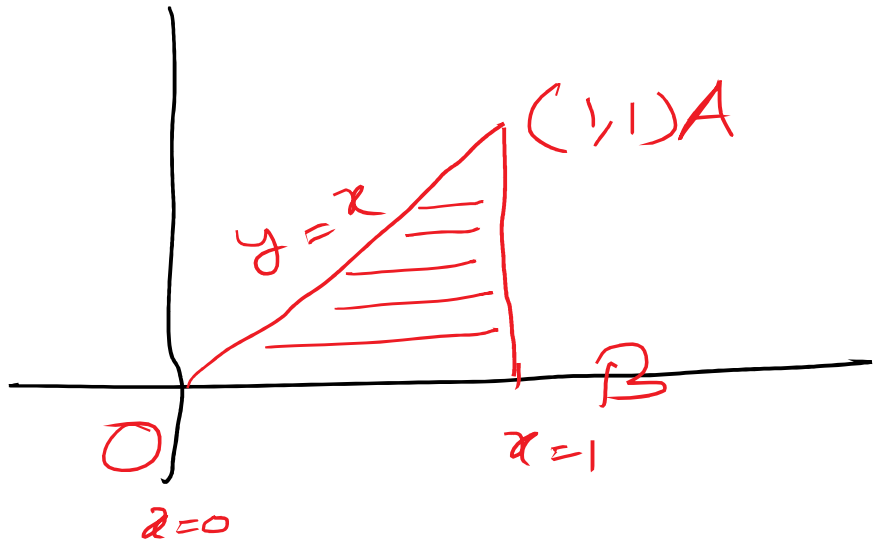
$$\int_{x_0}^{x_5} y\,dx = ?$$

Numerical Integral

Area of $\triangle OAB$

$= \frac{1}{2} \times \text{Base} \times \text{height}$

$= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$

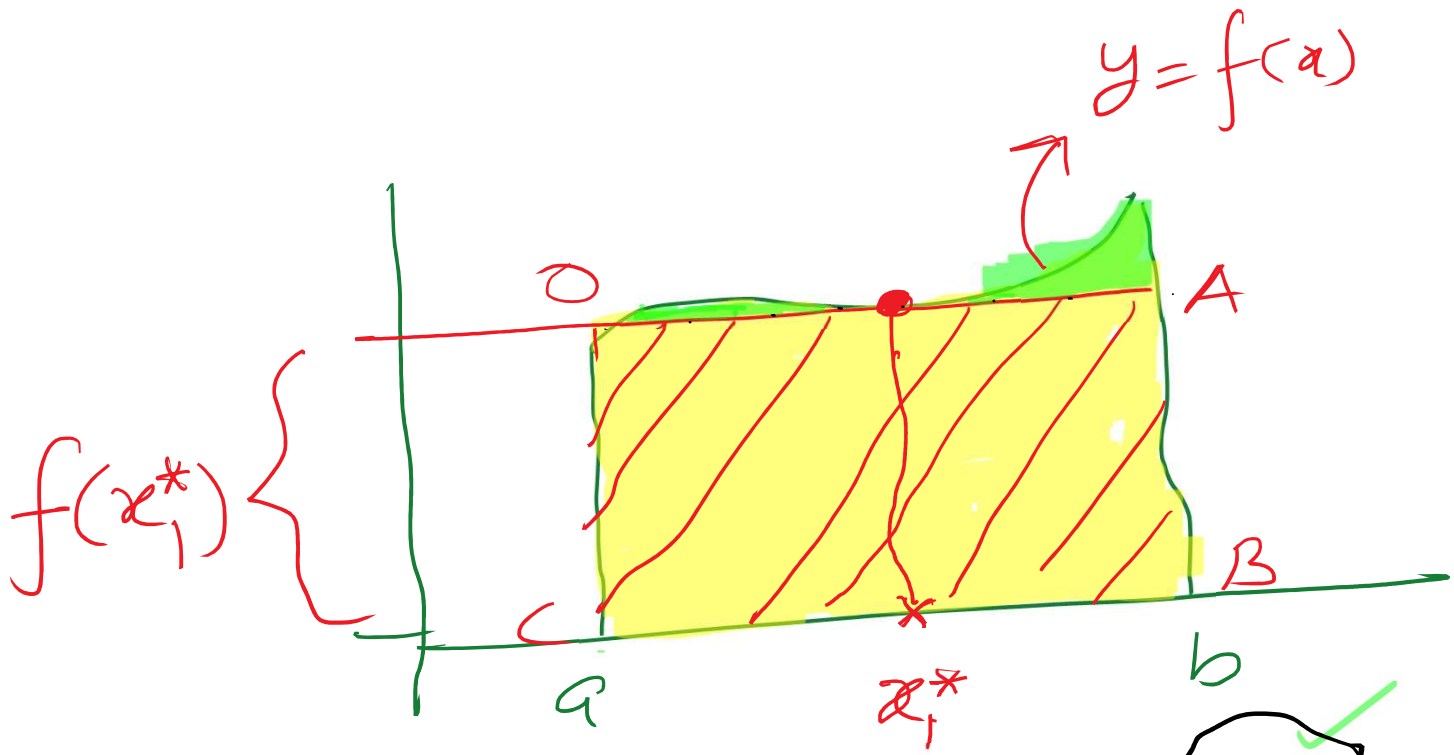


Area $\triangle OAB$

$$= \int_0^1 x\,dx = \frac{x^2}{2}\Big|_0^1 = \frac{1}{2}$$

Rectangular Method (Rect. Rule)

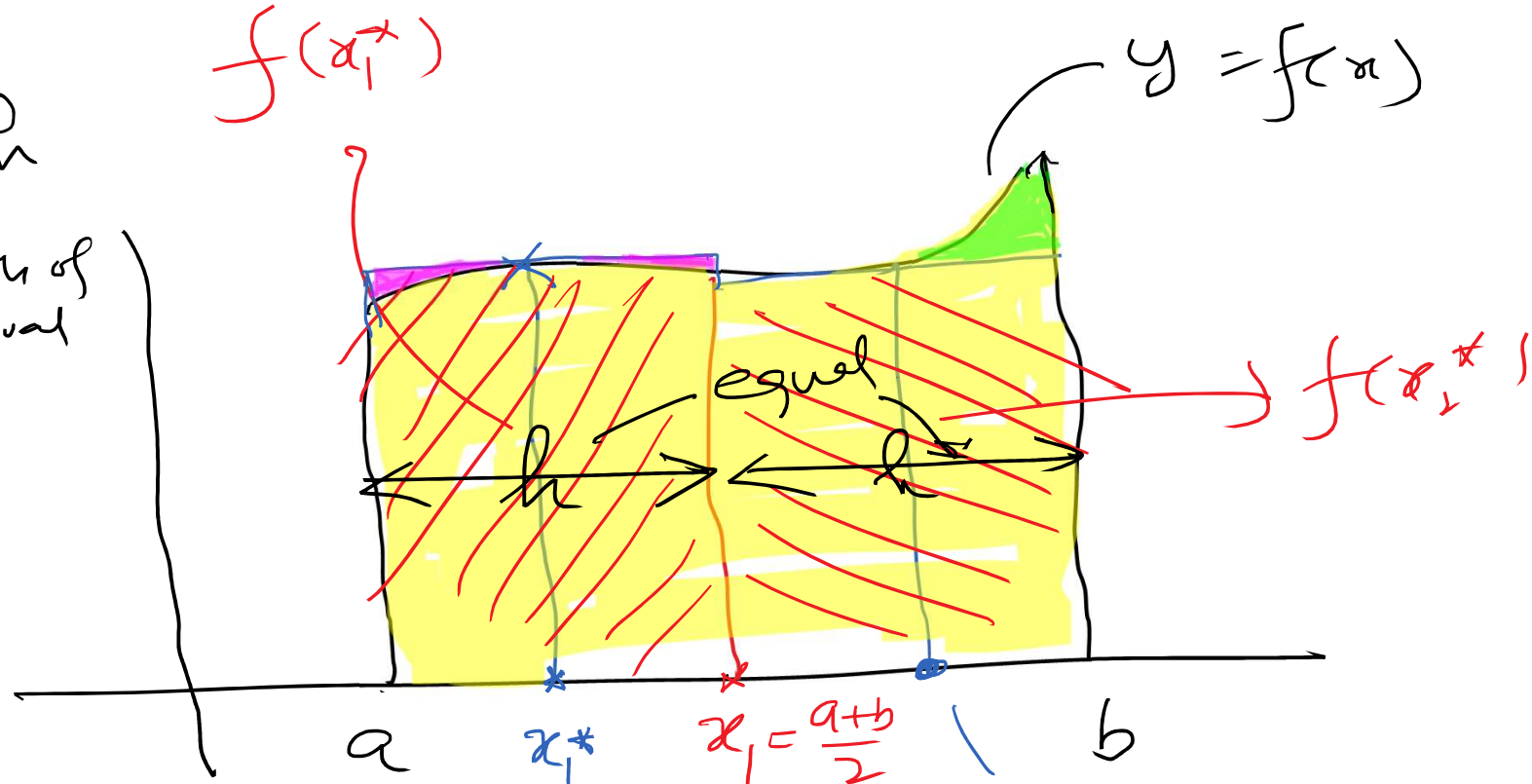
$$x_1^* = \frac{a+b}{2}$$



$$\text{Rect Area } OABC = (b-a) \times f(x_1^*) \approx \int_a^b f(x)\,dx$$

$$\frac{b-a}{2} = h$$

= width of interval



$$= \left(\frac{a + x_1}{2}\right)$$

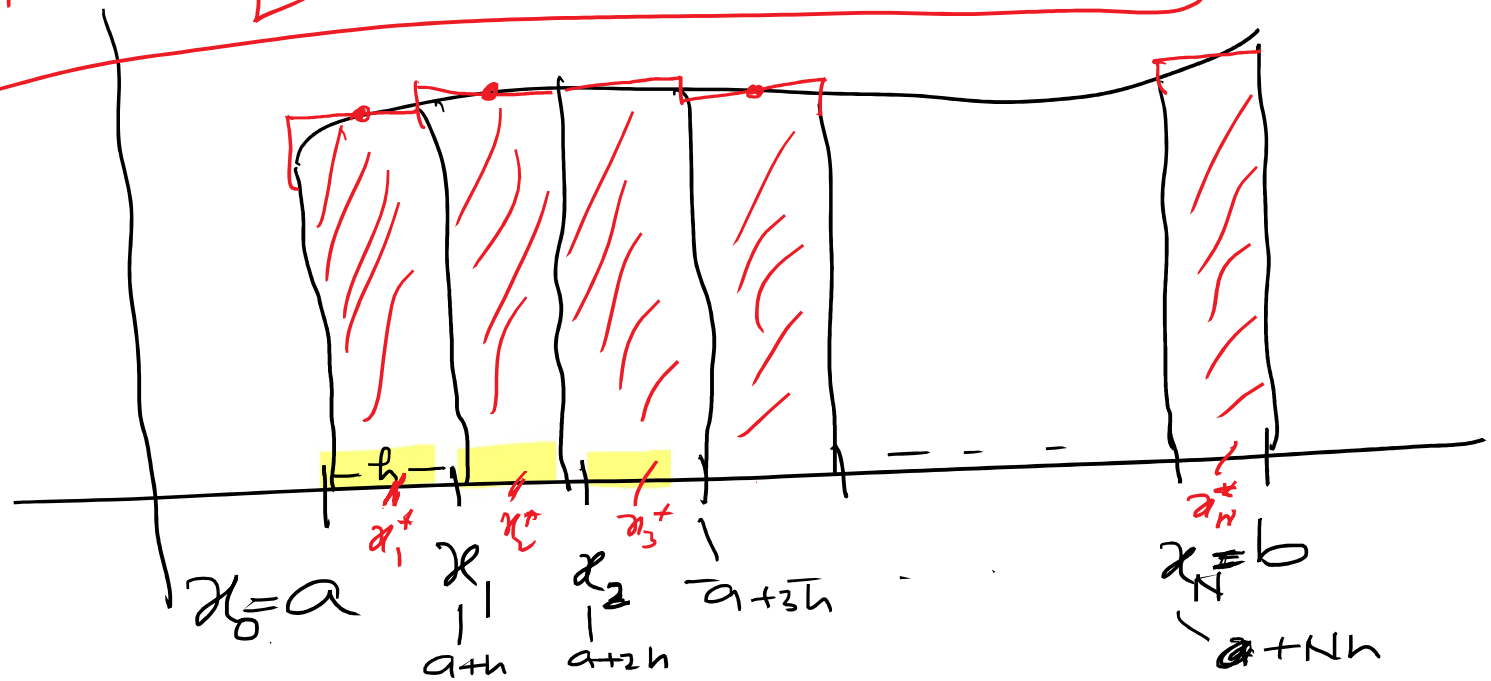
$$x_2^* = \frac{x_1 + b}{2}$$

$$h\left[f(x_1^*) + f(x_2^*)\right]$$

Sum of the two rectangles + green − purple $= \int_a^b f(x)\,dx$

$$(x_1 - a) \times f(x_1^*) + (b - x_1) \times f(x_2^*)$$

$$\int_a^b f(x)\,dx = h\left[f(x_1^*) + f(x_2^*) + \cdots + f(x_N^*)\right]$$



$$\frac{b-a}{N} = h$$

$$\Rightarrow \underline{\underline{b = a + Nh}}$$

$$x_1 = x_0 + h = a + h$$
$$x_2 = x_1 + h = a + h + h = a + 2h$$
$$x_3 = x_2 + h = a + 2h + h = a + 3h$$

$$x_j = a + jh$$

$$\int_0^1 x\,dx = ?$$

Rectangular Rule

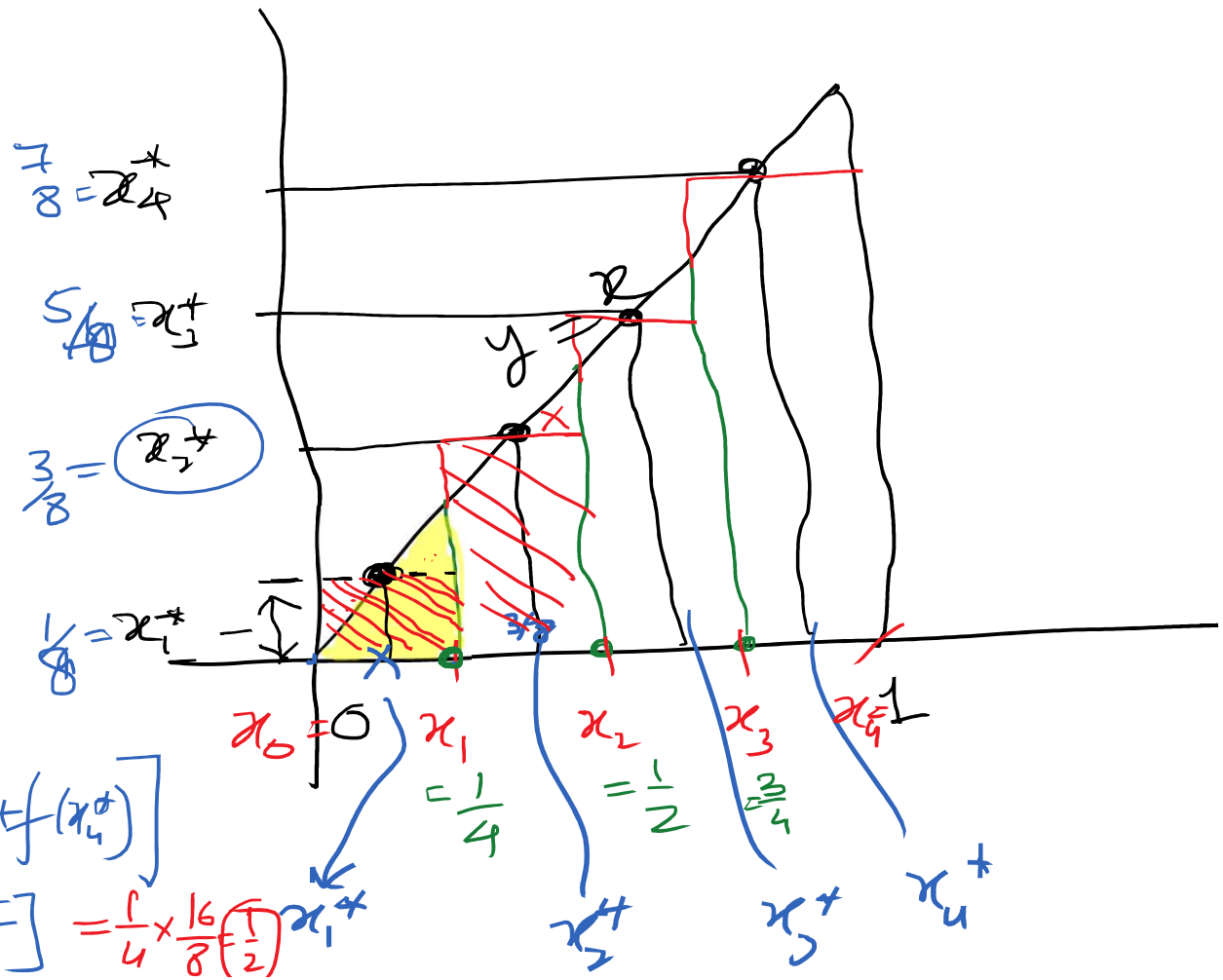
$$h = \frac{1-0}{4} = \frac{1}{4}$$

$$x_1^* = \frac{0+\frac{1}{4}}{2} = \frac{1}{8} \checkmark$$

$$x_2^* = \frac{\frac{1}{4}+\frac{1}{2}}{2} = \frac{3}{8}$$

$$x_3^* = \frac{\frac{1}{2}+\frac{3}{4}}{2} = \frac{5}{8}$$

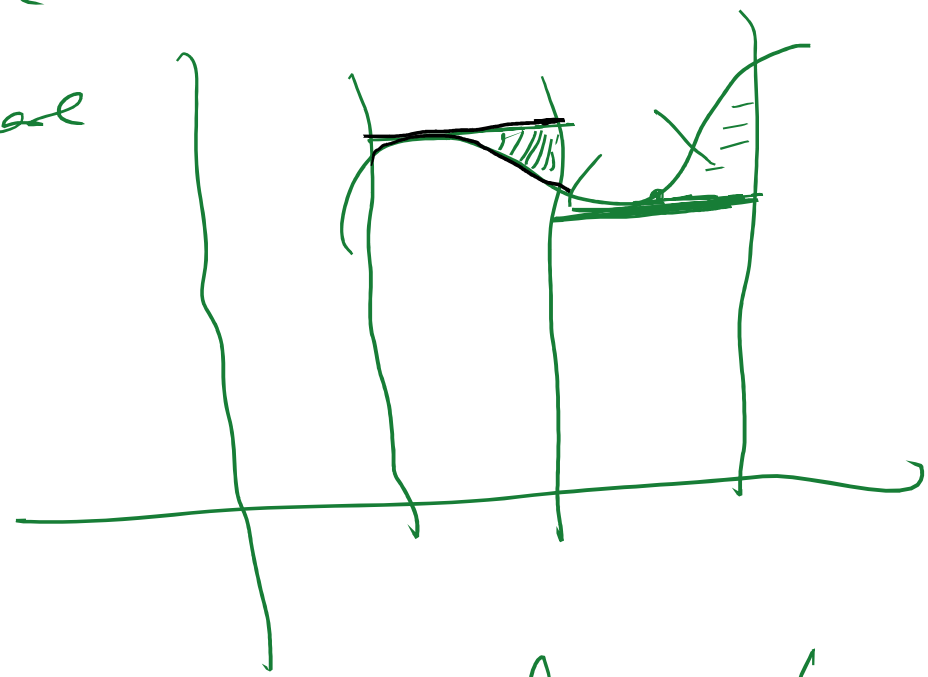
$$x_4^* = \frac{\frac{3}{4}+1}{2} = \frac{7}{8}$$



$$\int_0^1 x\,dx = h\left[f(x_1^*)+f(x_2^*)+f(x_3^*)+f(x_4^*)\right]$$

$$= \frac{1}{4}\left[\frac{1}{8}+\frac{3}{8}+\frac{5}{8}+\frac{7}{8}\right] = \frac{1}{4}\times\frac{16}{8} = \left(\frac{1}{2}\right)$$

If we increase N (decrease h) we shall get less error in $\int_a^b f(x)\,dx$ by rectangular rule.



In the rectangular rule

$f(x) = \text{constant}$

Newton-Cotes Methods/formulae

$[a,b] \longrightarrow$ Partition $[a,b]$ into n equal parts

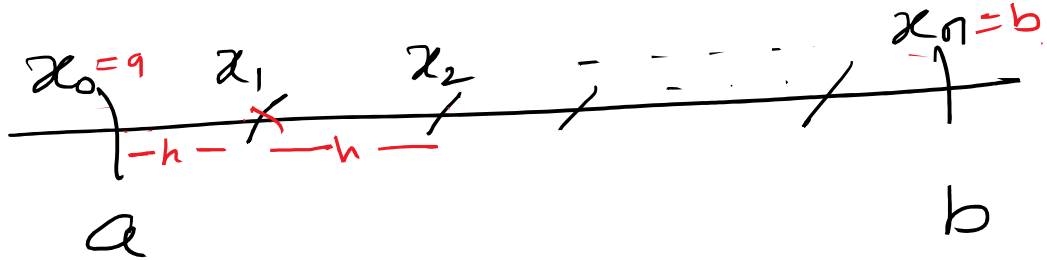
$h = \dfrac{b-a}{n}$, n number of partitions

Imp: $[a,b] \longrightarrow n$ parts
$\longrightarrow n+1$ points

$x_0 = a$
$x_1 = a+h$
$x_2 = a+2h$
$\vdots$
$x_j = a+jh$
$x_n = a+nh$

$\boxed{b = a+nh}$

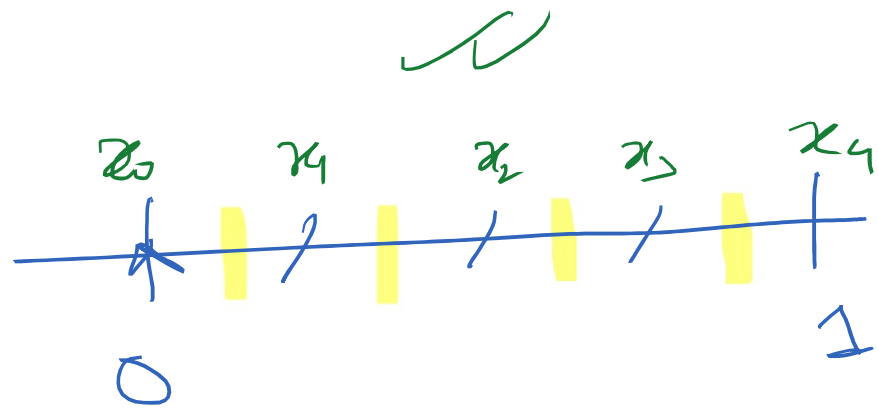
$$\boxed{\begin{array}{l} x_j = a+jh \\ 0 \le j \le n \end{array}}$$



$[0,1]$

$n = 4$

$h = \frac{1-0}{4} = \frac{1}{4} = \underline{0.25}$



partitions - 4

points - 5

$x_j = 0 + jh, \quad 0 \le j \le 4$

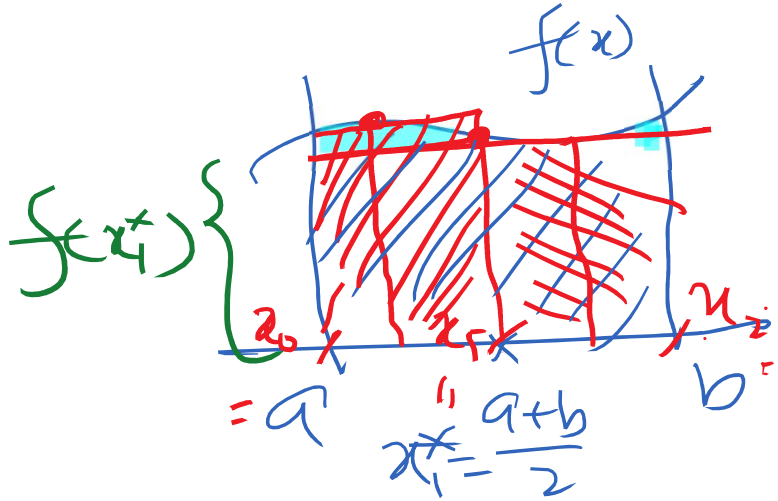
$$\left.\begin{aligned} x_0 &= 0 \\ x_1 &= 0.25 \\ x_2 &= 0.5 \\ x_3 &= 0.75 \\ x_4 &= 1 \end{aligned}\right\} \text{5 points}$$

→ 1) Rectangular Method (Rule)

$$I = \int_a^b f(x)\,dx$$

$$= \int_a^b f(x_1^*)\,dx$$

$$= f(x_1^*)\cdot(b-a)$$



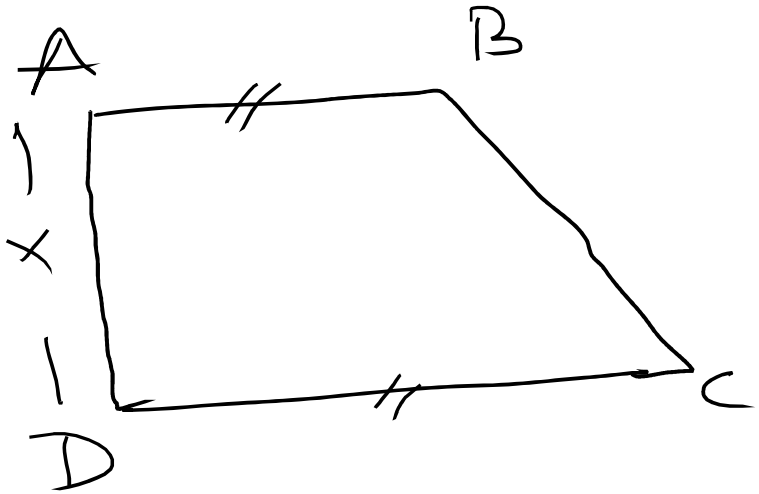
Error = Actual Area − Numerical Area

Imp: More partition will reduce error

2) Trapezoidal Method (Rule)

Trapezium ABCD

$$\text{Area} = \frac{AB + CD}{2} \times AD$$

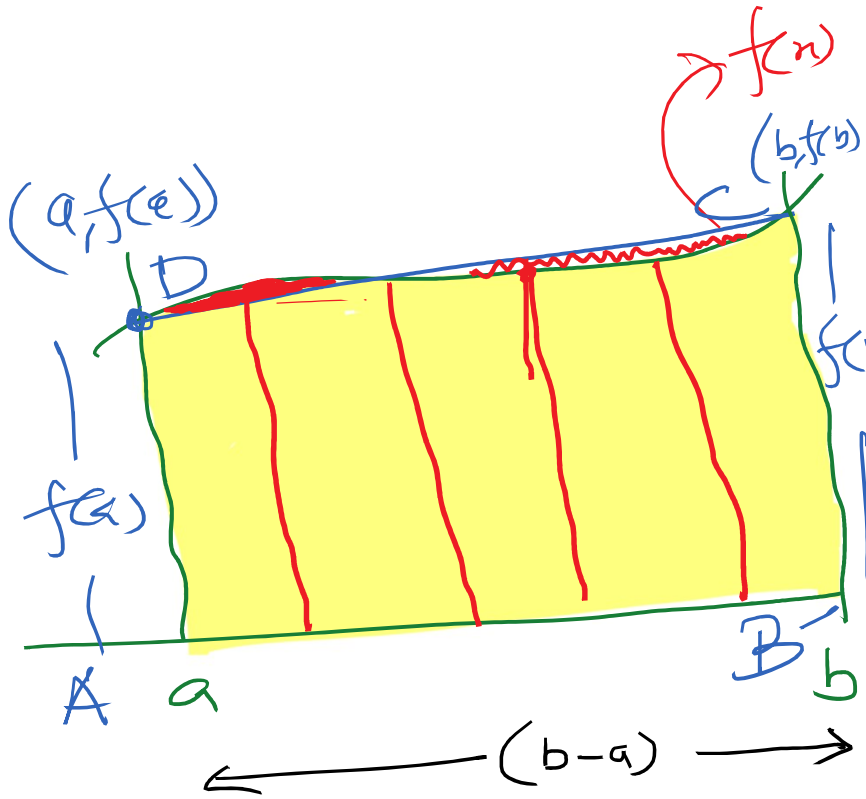


2) Trapezoidal Method

Area of Trapezium ABCD

$$\cong \int_a^b f(x)\,dx$$

$$\int_a^b f(x)\,dx \cong \frac{f(a)+f(b)}{2} \times (b-a)$$



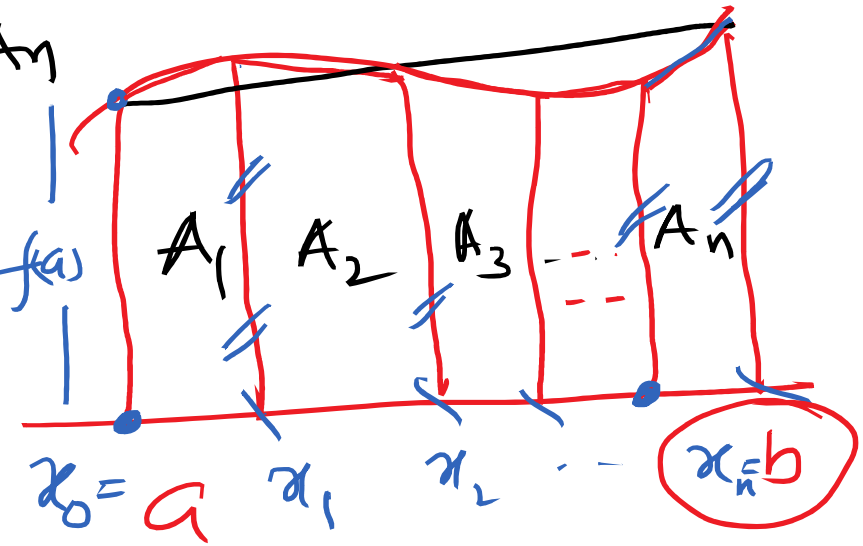
$h \rightarrow$ reduced

$\Rightarrow$ error is reduced.

$$\int_a^b f(x)\,dx \simeq A_1 + A_2 + A_3 + \cdots + A_n$$

$$= \left[\frac{f(a)+f(x_1)}{2} \times h\right] + \left[\frac{f(x_1)+f(x_2)}{2} \times h\right]$$

$$+ \cdots + \left[\frac{f(x_{n-1})+f(x_n)}{2} \times h\right]$$



$h = \frac{b-a}{n}$ length of each partition

$x_j = a + jh$

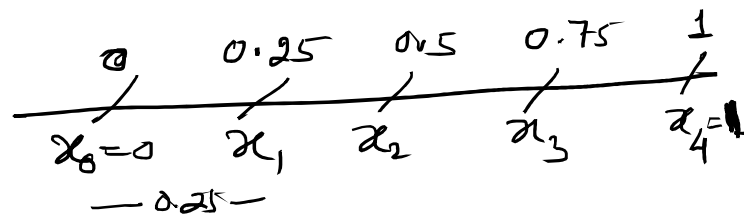
$0 \le j \le n$

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[f(a) + 2\left(f(x_1) + f(x_2) + \cdots + f(x_{n-1})\right) + f(b)\right]$$

Example Using Trapezoidal Method find $\int_0^1 e^{-x^2}\,dx = ?$

$a = 0,\ b = 1,\ f(x) = e^{-x^2}$ let $n = 4$

$h = \frac{1-0}{4} = 0.25$



$x_j = jh \qquad 0 \le j \le 4$

j	x_j	$e^{-x_j^2}$
0	$x_0 = 0$	$e^{-0} = 1 \longrightarrow f(x_0 = 0)$
1	$x_1 = 0.25$	$e^{-.25^2} = e^{-0.0625} = 0.9394 \longrightarrow f(x_1)$
2	$x_2 = 0.5$	$e^{-0.5^2} = e^{-0.25} = 0.7788 \longrightarrow f(x_2)$
3	$x_3 = 0.75$	$e^{-(0.75)^2} = 0.5697 \longrightarrow f(x_3)$
4	$x_4 = 1.0$	$e^{-1} = 0.3678 \longrightarrow f(x_4 = 1)$

$$\int_0^1 e^{-x^2}\,dx = \frac{h}{2}\left[f(a) + 2\left(f(x_1) + f(x_2) + f(x_3)\right) + f(b)\right]$$

$$= \frac{0.25}{2}\left[1 + 2\left(0.9394 + 0.7788 + 0.5697\right) + 0.3678\right]$$

$$= \underline{\underline{0.74295}}$$

Imp. for exam

1. How many decimal points should be considered ??

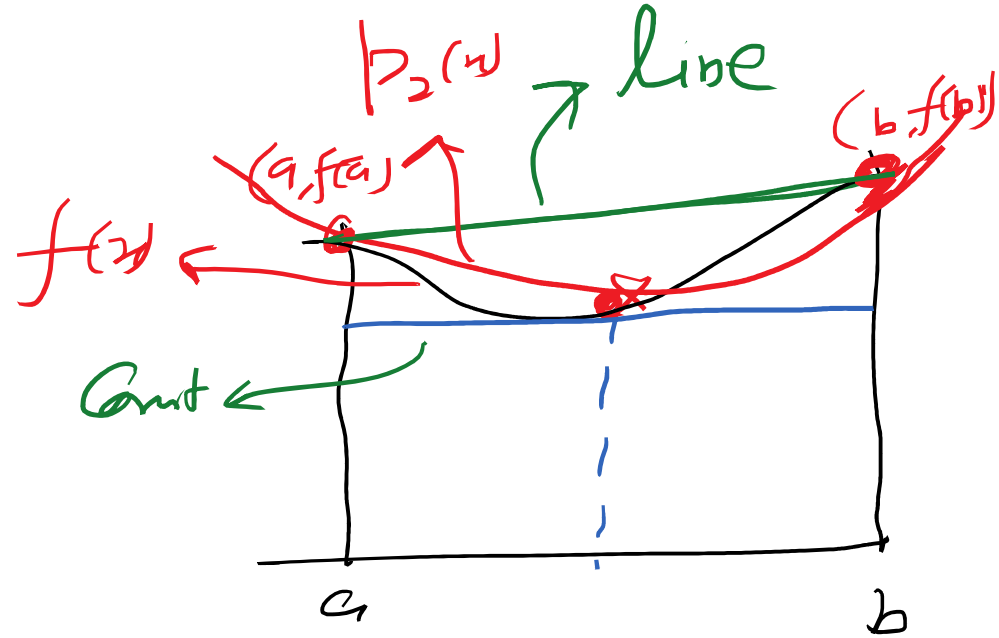
Ans Please do consider at Least 3 decimal points

2. Exam answers and your answer should match 2/3 decimal places

— CLOSEST one !!

— Rectangular Method

$f(x) \simeq$ Constant on $[a, b]$



— Trapezoidal

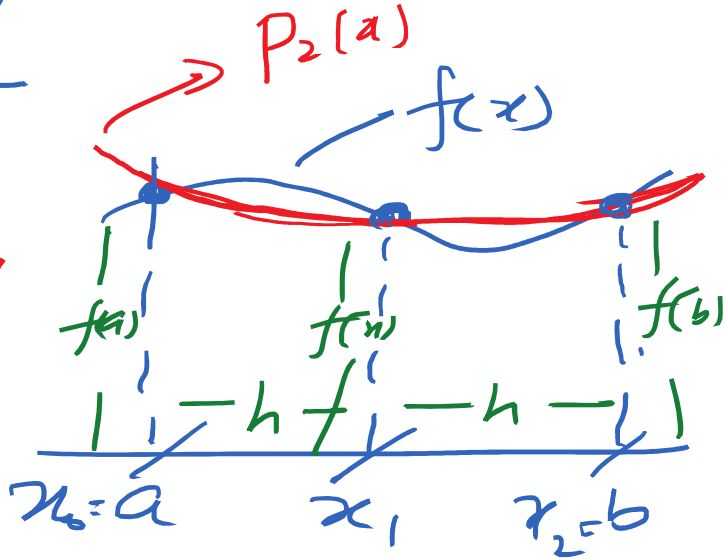
$f(x) \simeq$ Linear fun

$L(x) = \alpha + \beta x$

$f_2(x) = \alpha x^2 + \beta x + \gamma$

3) Simpson's $\frac{1}{3}$rd rule

$$P_2(x) = \alpha x^2 + \beta x + \gamma$$



$$\int_a^b f(x)\,dx = \int_a^b p_2(x)\,dx$$

[I leave it to next class]

2 partitions

$$x_1 = a + h$$

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[f(a) + 4f(x_1) + f(b)\right]$$

$$x_0 \longrightarrow f(x_0) \longrightarrow f_0$$

$$x_1 \longrightarrow f(x_1) \longrightarrow f_1$$

$$x_2 = f(x_2) \longrightarrow f_2$$

$$\vdots$$

$$x_j \longrightarrow f(x_j) \longrightarrow f_j$$

Ex Use Simpson's $\frac{1}{3}$rd rule to find

$$\int_0^1 e^{-x^2} = ?$$

Ans $n = 2$ $\quad a = 0,\ b = 1 \quad f(x) = e^{-x^2}$

$$h = \frac{1-0}{2} = 0.5$$

$x_0 = 0 \quad x_1 = 0.5,\ x_2 = 1$

$f_0 = 1 \quad f_1 = e^{-0.5^2} = e^{-0.25} = 0.7788$

$f_2 = e^{-1} = 0.3678$

$$\int_0^1 e^{-x^2}dx = \frac{h}{3}\left[f_0 + 4f_1 + f_2\right] = \frac{0.5}{3}\left[1 + 4\times 0.7788 + 0.3678\right]$$

$$= 0.7471$$

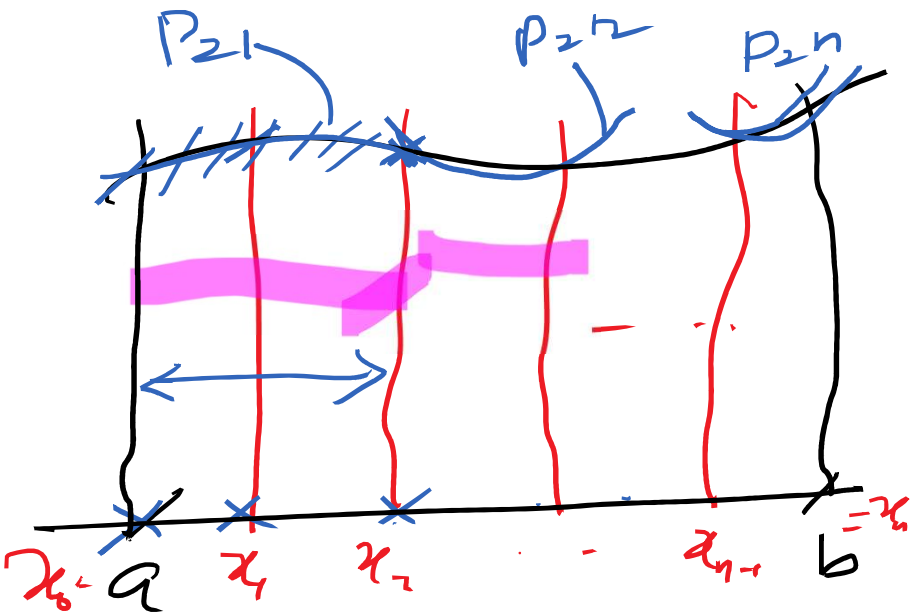
$$\int_a^b f(x)\,dx = \int_{x_0}^{x_2} P_{21}(x)\,dx$$

$$+ \int_{x_2}^{x_4} P_{22} + \int_{x_4}^{x_6} + \cdots + \int_{x_{n-2}}^{x_n}$$

$$= \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$

$$+ \frac{h}{3}\left[f(x_2) + 4f(x_3) + f(x_4)\right]$$

$$+ \cdots + \frac{h}{3}\left[f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\right]$$



$x_j = a + jh \quad 0 \le j \le n$

$$\boxed{n = 2 \times m}$$

Very Imp

$$\int_a^b f(x)\,dx = \frac{h}{3}\Big[\underbrace{f(x_0)}_{=a} + 2\big(f(x_2)+f(x_4)+\cdots+f(x_{n-2})\big)$$

$$+4\big(f(x_1)+f(x_3)+\cdots+f(x_{n-1})\big)+\underbrace{f(x_n)}_{=b}\Big]$$

or

$$\int_a^b f(x)\,dx = \frac{h}{3}\Big[f(a)+2\big(f(x_2)+f(x_4)+\cdots+f(x_{n-2})\big)$$

$$+4\big(f(x_1)+f(x_3)+\cdots+f(x_{n-1})\big)+f(b)$$

Ex Use Simpson's $\frac{1}{3}$rd rule $\int_0^1 e^{-x^2}\,dx = ?$ taking $n=10$

Ans $n = 10$ $\quad h = \frac{1-0}{10} = 0.1$ $\quad x_j = jh \quad 0 \le j \le 10$

j	x_j	$f_j = e^{-x_j^2}$
0	0	$e^{-0} = 1 \longrightarrow f_0$
1	0.1	$e^{-0.01} = 0.990050 \rightarrow f_1$
2	0.2	$0.960789 - f_2$
3	0.3	$e^{-0.09} = 0.913931 - f_3$
4	0.4	$e^{-0.16} = 852144 - f_4$
5	0.5	$e^{-0.25} = .976$ $0.778801 - f_5$
6	0.6	$e^{-0.36} = 0.697676 - f_6$
7	0.7	$0.612626 - f_7$
8	0.8	$0.527292 - f_8$
9	0.9	$0.444858 - f_9$
10	1.0	$0.367879 - f_{10}$

$f_0 = f(0)$, $f_{10} = f(1)$

$$\int_0^1 e^{-x^2}\,dx = \frac{0.1}{3}\left[f_0 + 4(f_1+f_3+f_5+f_7+f_9) + 2(f_2+f_4+f_6+f_8) + f_{10}\right]$$

$$= \frac{0.1}{3}\Big[1 + 4(0.990050 + \cdots + 0.444858)$$

$$+ 2(0.960789 + \cdots + 0.527292)$$

$$+ 0.367879\Big]$$

$=$ <u>Calculate yourself</u>

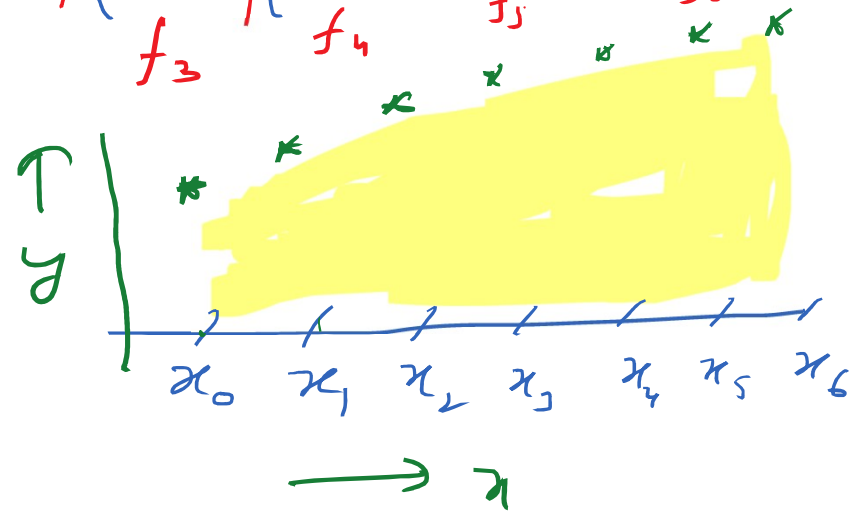
Example

Data

x	x_0 4	x_1 4.2	x_2 4.4	x_3 4.6	x_4 4.8	x_5 5	x_6 5.2
$f(x)$	1.3863 f_0	1.4351 f_1	1.4816 f_2	1.5261 f_3	1.5686 f_4	1.6094 f_5	1.6487 f_6

Find

$$\int_4^{5.2} f(x)\,dx = ?$$



Simpson's $\frac{1}{3}$rd rule

here $h = 0.2$ $n = 6$

$$\int_4^{5.2} f(x)\,dx = \frac{h}{3}\left[f_0 + 4(f_1+f_3+f_5) + 2(f_2+f_4) + f_6\right]$$

$$= \frac{0.2}{3}\left[\text{Substitute all the values}\right] = 1.82785$$

Example

Find $\int_0^2 \underbrace{\frac{\pi}{4} x^4 \cos\frac{\pi x}{4}}_{f(x)} dx = ?$

Simpson's $\frac{1}{3}$rd.

Case 1 Choose $h=1$ $n=2$ $x_1=1$

$$\int_0^2 f(x)dx = 0.740480$$

Ans.

$f_0=0$, $f_1=\frac{\pi}{4}\frac{1}{\sqrt{2}}$ $f_2=f(2)=0$

H.W.

$\boxed{n=4}$